

Legacy or Lineage? Framing the Future for sustainability

Gladys Barragan-Jason³, Gabriel Bayle¹, Cécile Bazart¹, Dimitri Dubois¹, Lisette Ibanez¹, Violette Pinçon¹, Sébastien Roussel^{1,2}, Arielle Syssau-Vaccarella² & Marc Willinger¹

¹CEE-M, Univ. Montpellier, CNRS, INRAe, Institut Agro

²EPSYLON, Université Paul-Valéry Montpellier 3, Montpellier, France

³CNRS, laboratoire SETE UAR 2029, Moulis, France

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Abstract

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How can we encourage present-day cooperation with future generations in managing common resources? We conduct a pre-registered experiment to test whether distinct psychological framings—legacy (future humanity), lineage (direct descendants), or perspective-taking (imagining oneself as future individuals)—differentially promote restraint in a common-pool resource dilemma. Compared to a baseline, emphasizing future generations in a concrete, temporal frame (“up to 2100”) significantly reduced extraction and increased cooperation. Surprisingly, framing the future in terms of one’s own descendants, or inviting participants to adopt a future generation’s perspective, did not further enhance cooperation. Our findings suggest that simple, universal framings of future impact—without invoking family ties or imaginative projection—are more effective at fostering sustainable behavior. These results contribute to the behavioral economics literature on intergenerational dilemmas by highlighting the role of temporal framing, rather than kinship or perspective-taking, in sustaining cooperation under intra-generational competition.

Keywords: Intergenerational cooperation; Common-pool resources; Behavioral nudges; Sustainability; Temporal framing; Social dilemmas; Experimental economics

1 Introduction

From climate change mitigation to biodiversity conservation, individuals routinely confront trade-offs between immediate private benefits and long-term collective welfare. Cooperation across generations poses a fundamental challenge in the governance of common-pool resources (CPRs) and public goods. These intergenerational settings can be seen as extensions of classic social dilemmas: while restraint today yields long-term efficiency gains, individuals face strong incentives to prioritize short-term self-interest (Ostrom, 1990; Hardin, 1968). However, intergenerational dilemmas differ from standard intra-generational ones in key respects: future individuals are absent from strategic interactions, cannot reciprocate, and lack mechanisms to enforce cooperation. Their welfare depends entirely on the

present generation’s preferences and decisions.

A large literature in economics and related fields has examined how cooperation is sustained within generations through repeated interactions, reputation, and formal institutions (Fehr Gächter, 2000; Chaudhuri, 2011; Dal Bó Fréchette, 2011). In such settings, cooperation is often stabilized by the threat of future sanctions or the promise of reciprocal benefits. In contrast, intergenerational dilemmas present a distinctive challenge: future individuals cannot reciprocate or punish, so cooperation hinges on the present generation’s intrinsic motivations or normative commitments. Models of intertemporal choice emphasize discounting as a key factor shaping individuals’ willingness to sacrifice present benefits for future gains (Laibson, 1997; Frederick et al., 2002).

Yet time preferences alone cannot fully explain cooperation failures, particularly when intra-generational competition amplifies short-term incentives to defect (Hauser et al., 2014; Tavoni et al., 2011). Recent work by Bayle et al. (2025, *Nature*) shows that cooperation with future generations declines sharply when individuals face intra-generational appropriation conflicts, even among those otherwise inclined to prioritize long-term welfare. If present-day conflicts undermine stewardship for the future, identifying interventions that sustain cooperation becomes all the more urgent.

While much experimental and theoretical work has examined mechanisms sustaining cooperation within generations—highlighting reciprocity, reputation, and enforcement (Fehr Gächter, 2000; Chaudhuri, 2011; Dal Bó Fréchette, 2011)—a smaller but growing literature addresses cooperation across generations, particularly in the context of environmental and resource management (IPCC, 2022; Hauser et al., 2014). These studies reveal how intra-generational conflict exacerbates intergenerational dilemmas, yet interventions to counteract these dynamics remain underexplored.

While behavioral interventions such as nudges have been widely studied as tools for aligning individual behavior with social welfare (Thaler Sunstein, 2008), their effectiveness in intergenerational contexts—where cooperation requires both present sacrifice and reasoning across temporal horizons—remains poorly understood. Behavioral economics has shown that small, strategically designed interventions can shift behavior in pro-social directions, especially when material incentives are weak or absent (Thaler Sunstein, 2008; Benartzi et al., 2017).

Nudges have proven effective in domains like environmental conservation (Schultz et al., 2007), financial planning (Madrian Shea, 2001), and charitable giving (Andreoni Serra-Garcia, 2022). However, their limits in addressing intergenerational trade-offs—where both immediate sacrifice and future-oriented reasoning are required—remain underexplored. Prior research suggests that individuals are more willing to incur costs for future others when they feel connected to them, whether through familial ties or through a perceived sense of continuity with their future selves (Hershfield et al., 2011; Bartels Rips, 2010).

Kin-based altruism draws on parochial preferences rooted in evolutionary psychology (Hamilton, 1964; Henrich Henrich, 2007), while universal moral appeals invoke broader notions of societal responsibility (Milfont et al., 2012). A growing body of research also

points to perspective-taking—the cognitive act of imagining oneself in another’s position—as a mechanism that expands moral concern and fosters future-oriented behavior (Batson, 2011; Gaesser, 2018). However, little is known about how these distinct mechanisms compare in promoting intergenerational cooperation, especially under the pressures of intra-generational competition.

We conducted a field experiment using a modified Intergenerational Goods Game (IGG). Participants make resource extraction decisions that trade off present consumption against benefits for future players, under experimental conditions designed to either foster or mitigate intra-generational competition. Specifically, we examined the effect of three types of nudge: (1) universal nudge that emphasize obligations to future generations, invoking concerns for societal continuity and welfare; (2) kinship-based nudge that presents future others as direct descendants, leveraging preferences for lineage preservation and familial continuity; and (3) perspective-taking nudge that prompt individuals to imagine themselves in the position of future individuals affected by present-day decisions.

In the current study, we contribute to the growing field on behavioral interventions and nudging, examining whether nudges targeting distinct psychological motivations can sustain cooperation when intra-generational rivalry is present. In doing so, our study advances understanding of cooperation in dynamic social dilemmas where present and future interests collide. It contributes to behavioral economics by probing the limits of nudging in environments characterized by both intra- and inter-generational conflicts. Moreover, it offers practical insights for designing institutions aimed at sustainability and long-term governance, where formal mechanisms might be complemented by behavioral interventions. Finally, our results speak to broader theories of social learning and norm transmission by shedding light on how cues about future others shape cooperative behavior.

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Preserving for the Future, Not Just Family

Who Do We Save For? Intergenerational vs. Kin-Based Nudges

Literature we need:

- Literature on sacrifice
- Literature on nudge and future design
- Literature on social learning
- Literature ICPR, CPR, PG Social dilemmas
- Cite Paper 1 to justify our choice of checking extractions instead of cooperation (because of sacrifice)

- Justifications for nudging in IPCC

2 Preregistered analyses

2.1 Extraction level per treatment

2.1.1 Main

Baseline vs each policy

Short vs shortkin (why?)

Same graph than Figure 2 of paper 1 for each treatment

2.1.2 Robustness

Cooperation level instead of extraction level

2.2 Survival rate per treatment

2.2.1 Main

Simulations with bootstrapping

2.2.2 Robustness

Analytical computation of survival rate

3 Exploratory analyses

3.1 Analysis by types

3.1.1 Main

Check the extractions per type and per treatment

Sacrifice 0-48; Egalitarian cooperator 49-50; Free-rider 51+

Check how many people are inconsistent (different types in different generations)

3.1.2 Robustness

Sacrifice 0-44; Egalitarian cooperator 45-50; Free-rider 51+

Sacrifice 0-49; Egalitarian cooperator 50; Free-rider 51+

3.2 Distribution of extraction decisions in sacrifice type

3.2.1 Main

Comparison of distribution of extraction per policy for the people in the sacrifice type

3.2.2 Robustness

Sacrifice 0-44; Egalitarian cooperator 45-50; Free-rider 51+

Sacrifice 0-49; Egalitarian cooperator 50; Free-rider 51+

3.3 Distribution of extraction decisions in free-rider type

3.3.1 Main

Comparison per policy

3.3.2 Robustness

Sacrifice 0-44; Egalitarian cooperator 45-50; Free-rider 51+

Sacrifice 0-49; Egalitarian cooperator 50; Free-rider 51+

3.4 Appendix

Explain why we don't take long treatment

References